

Math 113 – Lecture 04

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1 Cyclic Groups

Definition 1.1 (Cyclic). A group G is cyclic if

$$\exists a \in G \text{ such that } \langle a \rangle = G \text{ (where } \langle a \rangle \rightarrow \{\cdots, a^{-2}, a^{-1}, a^0 = e, a^1, \cdots\})$$

Theorem 1.2 (Commutativity) *Every Cyclic Group is Abelian.*

Proof. Let G be cyclic and $a \in G$ a generator of G .

Let $b, c \in G$, notice that $a^x = b, a^y = c$ for $x, y \in \mathbb{Z}$ WLOG assume $x \leq y$ Notice that

$$b * c = a^x * a^y = a^{x+y}$$

and

$$c * b = a^y * a^x = a^{y+x} = a^{x+y}$$

So commutativity holds and G is Abelian. □

Theorem 1.3 (Division in $\mathbb{Z}$) *If $n \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$ then*

$$\exists! q, r \in \mathbb{Z} \text{ such that } n = mq + r \text{ and } 0 \leq r < m$$

Remark 1.4 ($\exists!$). $\exists!$ denotes "there exists unique"

Theorem 1.5 (Subgroup Cyclic Properties) *Any subgroup of a cyclic group is cyclic*

Proof. G is cyclic and $\langle a \rangle = G$ and $H \leq G$.

If H is the identity, we have that $H = \langle e \rangle$ is cyclic.

If H is not the identity, it contains $k \neq e$. Since $\langle a \rangle = G$, we can write $k = a^n$ for some $n \in \mathbb{Z}$ with $n \neq 0$ (as $k \neq e$).

Defining the minimal exponent. Consider the set of positive exponents whose powers land in H :

$$S = \{j \in \mathbb{Z}^+ \mid a^j \in H\}.$$

S is nonempty: if $n > 0$ then $n \in S$ directly; if $n < 0$ then $a^{-n} = (a^n)^{-1} = k^{-1} \in H$, since H is a subgroup (closed under inverses), so $-n \in S$. By the well-ordering of $\mathbb{Z}^+$, S has a least element

$$m = \min S, \quad \text{i.e. } a^m \in H \text{ but } a^j \notin H \text{ for every } 0 < j < m.$$

This is the **minimality of m** — the one fact about m we exploit below.

Goal: show $H = \langle a^m \rangle$; then H is cyclic. The containment $\langle a^m \rangle \subseteq H$ is free: $a^m \in H$ and H is closed under products and inverses, so every power $(a^m)^q$, $q \in \mathbb{Z}$, stays in H (cyclic-subgroup theorem, Lecture 03).

For the other direction, take any $b \in H$. Since $H \subseteq G = \langle a \rangle$, we have $b = a^n$ for some $n \in \mathbb{Z}$, and it suffices to show $m \mid n$.

Use the division property: $n = qm + r$ with $0 \leq r < m$, and we want to show that $r = 0$. We get

$$a^n = (a^m)^q * a^r$$

Then inverse it up:

$$a^r = a^{n-qm} = (a^m)^{-q} * a^n$$

Both factors lie in H ($a^n = b \in H$, and $(a^m)^{-q} \in H$ by closure under inverses and products), so by closure $a^r \in H$.

Why $r = 0$ (the minimality step). We now know $a^r \in H$ together with $0 \leq r < m$. Suppose toward a contradiction that $r > 0$. Then $r \in \mathbb{Z}^+$, $r < m$, and $a^r \in H$ — that is, $r \in S$ with $r < m = \min S$, contradicting the minimality of m . So r cannot be positive, and $0 \leq r$ then forces $r = 0$.

Hence $m \mid n$, so $b = a^n = (a^m)^q \in \langle a^m \rangle$. Since $b \in H$ was arbitrary, $H \subseteq \langle a^m \rangle$, and therefore $H = \langle a^m \rangle$ is cyclic. $\square$

Corollary 1.6 (Scalars) *The subgroups of $\langle \mathbb{Z}, + \rangle$ are the groups $\langle n\mathbb{Z}, + \rangle$ for $n \in \mathbb{Z}_{\geq 0}$*

Proof. $\langle \mathbb{Z}, + \rangle$ is cyclic, generated by 1: in additive notation the powers a^j become multiples $j \cdot 1$, and every $z \in \mathbb{Z}$ is $z = z \cdot 1$. So the theorem above applies with $a = 1$, reading " a^j " as " j ".

Let $H \leq \langle \mathbb{Z}, + \rangle$. If $H = \{0\}$, then $H = 0\mathbb{Z}$ and we are done. Otherwise H contains some $k \neq 0$; if $k < 0$ then $-k \in H$ (inverses), so H contains a positive integer and, by well-ordering, a least positive one:

$$m = \min\{j \in \mathbb{Z}^+ \mid j \in H\}.$$

We claim $H = m\mathbb{Z}$. Since $m \in H$ and H is closed under addition and negation, $qm \in H$ for every $q \in \mathbb{Z}$, so $m\mathbb{Z} \subseteq H$. Conversely, let $h \in H$ and divide: $h = qm + r$ with $0 \leq r < m$. Then

$$r = h - qm \in H$$

by closure under differences, so the minimality of m forces $r = 0$ exactly as in the theorem above. Hence $m \mid h$, i.e. $h \in m\mathbb{Z}$, giving $H = m\mathbb{Z} = \langle m\mathbb{Z}, + \rangle$.

Conversely, every $n\mathbb{Z}$ with $n \in \mathbb{Z}_{\geq 0}$ really is a subgroup: $0 = n \cdot 0 \in n\mathbb{Z}$, and $nx - ny = n(x - y) \in n\mathbb{Z}$.

So the subgroups of $\langle \mathbb{Z}, + \rangle$ are exactly the $\langle n\mathbb{Z}, + \rangle$ for $n \in \mathbb{Z}_{\geq 0}$. $\square$

Theorem 1.7 (GCD) *If $r, s \in \mathbb{Z}^+$, then $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\langle \mathbb{Z}, + \rangle$.*

Proof. First, $H \neq \emptyset$: $0 = 0 \cdot r + 0 \cdot s \in H$ (and note $r = 1 \cdot r + 0 \cdot s \in H$ and $s = 0 \cdot r + 1 \cdot s \in H$).

Now check closure under differences (the one-step subgroup test). Take $h_1 = n_1r + m_1s$ and $h_2 = n_2r + m_2s$ in H , with $n_i, m_i \in \mathbb{Z}$. Then

$$h_1 - h_2 = (n_1 - n_2)r + (m_1 - m_2)s$$

which is again an integer multiple of r plus an integer multiple of s , so $h_1 - h_2 \in H$. A nonempty subset of a group that is closed under xy^{-1} (here, under $x - y$) is a subgroup, so $H \leq \langle \mathbb{Z}, + \rangle$. $\square$

Remark 1.8 (What is H ?). We will in fact have that $H = d\mathbb{Z}$ where $d = \gcd(r, s)$:

By the Scalars corollary, $H = n\mathbb{Z}$ for some $n \in \mathbb{Z}_{\geq 0}$. Since $r \in H$ and $r > 0$, we must have $n \neq 0$; write $d = n \in \mathbb{Z}^+$. Because $r, s \in H = d\mathbb{Z}$, both are multiples of d .

Also $d \in d\mathbb{Z} = H$, so d itself has the form $d = nr + ms$ for some $n, m \in \mathbb{Z}$ — a Bézout identity for the gcd.

Proposition 1.9 (What is D ?) *d is a divisor of both r and s , and any other such divisor d' must divide d (so d is the greatest common divisor, $d = \gcd(r, s)$).*

Proof. We already showed $d \mid r$ and $d \mid s$: both r and s lie in $d\mathbb{Z} = H$.

Let d' be any common divisor of r and s , so $r = d'x$ and $s = d'y$ for some $x, y \in \mathbb{Z}$. Feed these into the Bézout form of d from the remark:

$$d = nr + ms = n(d'x) + m(d'y) = d'(nx + my),$$

so $d' \mid d$. Every common divisor of r and s divides d ; in particular no common divisor is larger than d , which is exactly what makes d the greatest common divisor. $\square$

Definition 1.10 (Order of an element). The order of $a \in G$, written $\text{ord}(a)$, is the least $n \in \mathbb{Z}^+$ with $a^n = e$, if such an n exists; otherwise a has infinite order.

The "least" is the same minimality move as before: if $a^k = e$ for any $k \neq 0$, then also $a^{-k} = e$, so $\{j \in \mathbb{Z}^+ \mid a^j = e\}$ is nonempty and well-ordering produces the least element.

Theorem 1.11 (Classification of Cyclic Groups) *Let $G = \langle a \rangle$ be cyclic. If a has infinite order, then $G \cong \langle \mathbb{Z}, + \rangle$; if $\text{ord}(a) = n < \infty$, then $G \cong \langle \mathbb{Z}_n, + \rangle$.*

Proof. **Case 1: infinite order.** Define

$$\varphi : \mathbb{Z} \rightarrow G, \quad \varphi(k) = a^k.$$

Homomorphism: $\varphi(j+k) = a^{j+k} = a^j * a^k = \varphi(j) * \varphi(k)$ (exponent law).

Surjective: every element of $G = \langle a \rangle$ is some $a^k = \varphi(k)$.

Injective: suppose $\varphi(j) = \varphi(k)$ with $j \neq k$, WLOG $j > k$. Then

$$a^{j-k} = a^j * (a^k)^{-1} = e$$

with $j-k \in \mathbb{Z}^+$, contradicting infinite order (no positive power of a is e). So φ is an isomorphism and $G \cong \langle \mathbb{Z}, + \rangle$.

Case 2: $\text{ord}(a) = n < \infty$. The key fact is

$$a^i = a^j \iff i \equiv j \pmod{n}.$$

($\Leftarrow$): if $i = j + qn$, then $a^i = a^j * (a^n)^q = a^j * e = a^j$.

($\Rightarrow$): if $a^i = a^j$, then $a^{i-j} = a^i * (a^j)^{-1} = e$. Divide (the division property applies to any integer): $i-j = qn + r$ with $0 \leq r < n$. Then

$$a^r = a^{i-j} * (a^n)^{-q} = e,$$

and minimality of n (the least positive exponent with power e) forces $r = 0$, exactly as in the subgroup proof. So $n \mid i-j$, i.e. $i \equiv j \pmod{n}$.

Now define

$$\varphi : \mathbb{Z}_n \rightarrow G, \quad \varphi([k]) = a^k.$$

The thing to check for a map out of $\mathbb{Z}_n$ is that it is well-defined: if $[k] = [k']$ then $k \equiv k' \pmod{n}$, so $a^k = a^{k'}$ by ($\Leftarrow$), and φ does not depend on the representative.

Homomorphism: $\varphi([j] + [k]) = \varphi([j+k]) = a^{j+k} = a^j * a^k = \varphi([j]) * \varphi([k])$.

Surjective as before, and injective by ($\Rightarrow$): $\varphi([j]) = \varphi([k])$ gives $j \equiv k \pmod{n}$, i.e. $[j] = [k]$.

So $G \cong \langle \mathbb{Z}_n, + \rangle$; in particular the distinct elements of G are exactly $e, a, \dots, a^{n-1}$, so $|G| = n = \text{ord}(a)$. $\square$

Theorem 1.12 (Order of a Power) *Let $G = \langle a \rangle$ be cyclic with $\text{ord}(a) = n$. Then for any $s \in \mathbb{Z}^+$, the subgroup $\langle a^s \rangle$ has order*

$$\text{ord}(a^s) = |\langle a^s \rangle| = \frac{n}{\gcd(s, n)}.$$

Proof. Write $d = \gcd(s, n)$ and factor $s = ds'$, $n = dn'$. Then $\gcd(s', n') = 1$: any common divisor of s' and n' would turn d into a larger common divisor of s and n .

From the classification proof we have the key fact (take $j = 0$ in $a^i = a^j \iff i \equiv j \pmod{n}$):

$$a^j = e \iff n \mid j.$$

Now $(a^s)^k = a^{sk} = e \iff n \mid sk$. Divide through by d :

$$n \mid sk \iff dn' \mid ds'k \iff n' \mid s'k.$$

Since $\gcd(s', n') = 1$, Bézout (the GCD remark above) gives $x, y \in \mathbb{Z}$ with $xs' + yn' = 1$. Multiplying by k :

$$k = xs'k + yn'k,$$

and if $n' \mid s'k$ then n' divides both terms on the right, so $n' \mid k$; conversely $n' \mid k$ trivially gives $n' \mid s'k$. Hence

$$(a^s)^k = e \iff n' \mid k.$$

So the exponents that kill a^s are exactly the multiples of $n' = n/d$, and the least positive one — which is $\text{ord}(a^s)$ by definition, picked out by well-ordering as always — is

$$\text{ord}(a^s) = n' = \frac{n}{d} = \frac{n}{\gcd(s, n)}. \quad \blacksquare$$

□

Theorem 1.13 (Same Subgroup from Two Powers) *Let $G = \langle a \rangle$ be cyclic with $\text{ord}(a) = n$. Then for $s, t \in \mathbb{Z}^+$,*

$$\langle a^s \rangle = \langle a^t \rangle \iff \gcd(s, n) = \gcd(t, n).$$

Proof. ($\Leftarrow$) Suppose $\gcd(s, n) = \gcd(t, n) = d$. We first claim

$$\langle a^s \rangle = \langle a^d \rangle$$

(and likewise $\langle a^t \rangle = \langle a^d \rangle$, which finishes this direction).

$\langle a^s \rangle \subseteq \langle a^d \rangle$: since $d \mid s$, write $s = ds'$; then $a^s = (a^d)^{s'} \in \langle a^d \rangle$, and $\langle a^s \rangle$ is the smallest subgroup containing a^s (cyclic-subgroup theorem, Lecture 03).

$\langle a^d \rangle \subseteq \langle a^s \rangle$: by Bézout, $d = xs + yn$ for some $x, y \in \mathbb{Z}$, so

$$a^d = a^{xs+yn} = (a^s)^x * (a^n)^y = (a^s)^x * e = (a^s)^x \in \langle a^s \rangle.$$

($\Rightarrow$) Suppose $\langle a^s \rangle = \langle a^t \rangle$. Equal sets have equal cardinality, so by the Order of a Power theorem

$$\frac{n}{\gcd(s, n)} = |\langle a^s \rangle| = |\langle a^t \rangle| = \frac{n}{\gcd(t, n)},$$

and canceling n : $\gcd(s, n) = \gcd(t, n)$. □

Corollary 1.14 (Subgroups of a Finite Cyclic Group) *A cyclic group of order n has exactly one subgroup of each order d dividing n — and no other subgroups. The subgroup of order d is $\langle a^{n/d} \rangle$.*

Proof. Every $H \leq \langle a \rangle$ is $\langle a^m \rangle$ for some $m \in \mathbb{Z}^+$ (subgroup theorem; $\{e\} = \langle a^n \rangle$), and by the ($\Leftarrow$) direction $\langle a^m \rangle = \langle a^{\gcd(m, n)} \rangle$, which has order $n/\gcd(m, n)$ where $\gcd(m, n)$ divides n . As the divisor

$d' = \gcd(m, n)$ runs over all divisors of n , so does the order n/d' : every subgroup is $\langle a^{d'} \rangle$ for some $d' \mid n$, of order n/d' , and the subgroup of order d is $\langle a^{n/d} \rangle$ (since $\gcd(n/d, n) = n/d$, its order is $n/(n/d) = d$).

Uniqueness: if $\langle a^{d_1} \rangle = \langle a^{d_2} \rangle$ with $d_1, d_2 \mid n$, then by the $(\Rightarrow)$ direction $\gcd(d_1, n) = \gcd(d_2, n)$, i.e. $d_1 = d_2$. $\square$

Example 1.15 (The Complete Picture for $\text{ord}(a) = 12$). Let $\text{ord}(a) = 12$. The two theorems give the entire subgroup lattice at a glance:

s	$\gcd(s, 12)$	$\text{ord}(a^s) = \frac{12}{\gcd(s, 12)}$	$\langle a^s \rangle$
1, 5, 7, 11	1	12	G itself
2, 10	2	6	$\{e, a^2, a^4, \dots, a^{10}\}$
3, 9	3	4	$\{e, a^3, a^6, a^9\}$
4, 8	4	3	$\{e, a^4, a^8\}$
6	6	2	$\{e, a^6\}$
12	12	1	$\{e\}$

Each row is a single subgroup — six subgroups for the six divisors of 12, exactly as the corollary predicts — and two powers generate the same subgroup precisely when they sit in the same row, i.e. share their $\gcd$ with 12.

Reading the first row: a^s is a **generator** of G iff $\gcd(s, 12) = 1$, so $G = \langle a \rangle = \langle a^5 \rangle = \langle a^7 \rangle = \langle a^{11} \rangle$ — all four generators.

Example 1.16 ($\langle a^8 \rangle = \langle a^4 \rangle$ in Detail). Still with $n = 12$: since $\gcd(8, 12) = 4 = \gcd(4, 12)$, the theorem says $\langle a^8 \rangle = \langle a^4 \rangle$. Check it by hand:

$$\langle a^8 \rangle = \{e, a^8, (a^8)^2\} = \{e, a^8, a^{16}\} = \{e, a^8, a^4\},$$

using $a^{16} = a^4$ because $16 \equiv 4 \pmod{12}$; and next $(a^8)^3 = a^{24} = (a^{12})^2 = e$ stops the list. So both subgroups are $\{e, a^4, a^8\}$, of order $12/\gcd(\cdot, 12) = 12/4 = 3$, and no smaller power of a^8 gives e ($a^8 \neq e$ and $(a^8)^2 = a^4 \neq e$) — the formula $\text{ord}(a^8) = 3$ confirmed directly.

In additive notation, i.e. inside $\langle \mathbb{Z}_{12}, + \rangle$ with $a \leftrightarrow [1]$ (via the classification theorem): $\langle [8] \rangle = \{[0], [8], [4]\} = \langle [4] \rangle$ — repeatedly adding 8 mod 12 walks $[0] \rightarrow [8] \rightarrow [4] \rightarrow [0]$, the same triangle as adding 4, just traversed in a different order.

2 Definitions and notation

Definition 2.1 (First definition). Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: